

# Project Notes & Components List

## ① Components Required

### i) ESP32 Devkit

Use = Main MCU.

### ii) ESP32 Cam : + SD card holder

Use : Camera for video Recording and Transmission via wifi to Laptop for Computer Vision.

### iii) L298N :

Use: Motor Driver.

### iv) Motors :

Yellow DC Motors or N20 high (1000rpm) RPM Motors

### (v) MPU6050 :

Use: Inertial Measurement unit , heading or Position estimation .

① - gyroscope → Rough heading

+

- motor run time

⇒

**" Estimated Explored or UnExplored Area "**

② In Case The robot Tilted at some Position , Accelerometer will give the data ( E.g., If Robot becomes upside down , instead of waiting for connection Time-out , It will Send a Message To Laptop.

### vi) Ultrasonic sensors :

Use: For precise obstacle detection (Because computer vision model will only be trained for seeing humans not walls for simplicity).

### vii) Power & Battery System :

- 2x 18650 cells
- Battery holder
- Tp5100 charging module
- 2s BMS

### viii) Misc:

wires, header pins etc.

### ix) Chasis :

I'd suggest to design & make our own chasis because Everything Else Inc. Electronics is common and Software is Not Included for us , So only Thing That could be attractive and different is Custom 3d printed design.

### x) Thermal Camera (Not Recommended) :

I will not suggest To use Thermal Camera because of its cost ( if we take a better one approx. 80 \$ for ESP 32 S3 IR Module ) and low range ( hardly 1m ) and low Efficiency ( only blob of hot objects . It can lead To false detection ) If we use cheaper one . So This should be abstained.

## Detection ways

↳ Computer Vision

↳ Distorted wifi Signals .

( That is obtained using a wifi router by Connecting Every Components wifi To it )

*\* Note: WiFi signals can pass through lean walls so it also lacks the element of wall and obstacle detection.*

### wifi router

↳ Can be mobile etc ( Its Not an issue ) .

## ② Parts Available & Required

*Original Note: "Every part is available Except MPU6050 and ESP32 Cam."*

### Available Parts:

- ESP32 Devkit
- L298N Motor Driver
- Motors (Yellow DC Motors / N20 high 1000rpm)
- 2x 18650 cells
- Misc (wires, header pins, etc.)

### Required Parts (Needed to Buy):

- MPU6050
- ESP32 Cam (+ SD card holder)
- Ultrasonic sensors
- Battery holder
- Tp5100 charging module
- 2s BMS